# Mobile Application Development

## Practical No. 14

## Design and develop any convertor application like temperature convertor /currency convertor/ volume convertor.

### 1 – Design a Currency Convertor:-

- **activity_main.xml :-**

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">


    <!-- Title -->

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Currency Converter"
        android:textSize="24sp"
        android:textStyle="bold"
        android:paddingBottom="20dp"/>


    <!-- Input Amount -->

    <EditText
        android:id="@+id/editAmount"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Amount"
        android:inputType="numberDecimal"
        android:padding="10dp"/>


    <!-- USD to INR Button -->

    <Button
        android:id="@+id/btnUsdToInr"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="USD to INR"
        android:layout_marginTop="10dp"/>
```

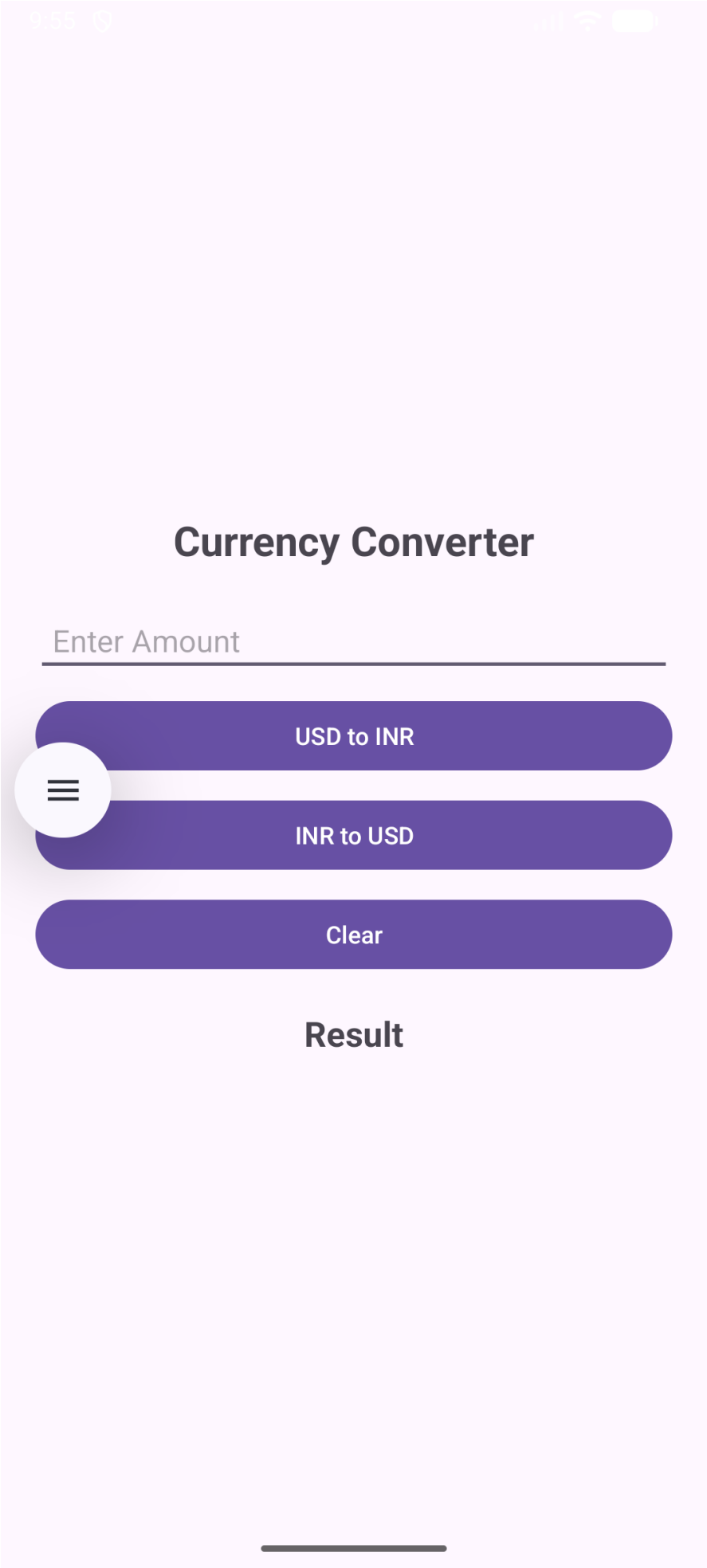
```
    <!-- INR to USD Button -->

    <Button
        android:id="@+id/btnInrToUsd"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="INR to USD"
        android:layout_marginTop="10dp"/>


    <Button
        android:id="@+id/btnClear"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Clear"
        android:layout_marginTop="10dp"/>


    <!-- Result -->

    <TextView
        android:id="@+id/textResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"/>


</LinearLayout>
```

- **MainActivity.java**

```
package com.example.practical14;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText editAmount;
    TextView textResult;

    double rate = 83.0;   // 1 USD = 83 INR

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        editAmount = findViewById(R.id.editAmount);
        textResult = findViewById(R.id.textResult);

        Button btnUsdToInr = findViewById(R.id.btnUsdToInr);
        Button btnInrToUsd = findViewById(R.id.btnInrToUsd);
        Button btnClear = findViewById(R.id.btnClear);   // Clear Button

        btnUsdToInr.setOnClickListener(v -> convertUsdToInr());
        btnInrToUsd.setOnClickListener(v -> convertInrToUsd());

        // Clear Button Action
        btnClear.setOnClickListener(v -> {
            editAmount.setText("");
            textResult.setText("Result");
        });
    }

    private void convertUsdToInr() {
        String value = editAmount.getText().toString();

        if (value.isEmpty()) {
            Toast.makeText(this, "Enter amount",
Toast.LENGTH_SHORT).show();
            return;
        }

        double usd = Double.parseDouble(value);
        double inr = usd * rate;

        textResult.setText("₹ " + inr);
    }

    private void convertInrToUsd() {
        String value = editAmount.getText().toString();

        if (value.isEmpty()) {
            Toast.makeText(this, "Enter amount",
Toast.LENGTH_SHORT).show();
            return;
        }

        double inr = Double.parseDouble(value);
        double usd = inr / rate;

        textResult.setText("$ " + usd);
    }
}
```

- **Output:-**


9:55
Currency Converter
Enter Amount
USD to INR
INR to USD
Clear
Result